

N. GREGORY

MANKIW

PRINCIPLES OF

ECONOMICS

Eighth Edition



CHAPTER

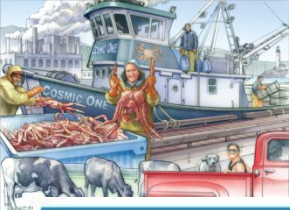
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Externalities

Premium PowerPoint Slides by:
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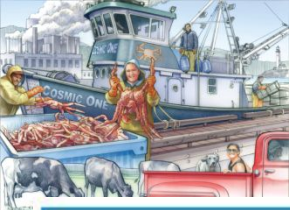
Look for the answers to these questions:

- What is an externality?
- Why do externalities make market outcomes inefficient?
- What public policies aim to solve the problem of externalities?
- How can people sometimes solve the problem of externalities on their own? Why do such private solutions not always work?



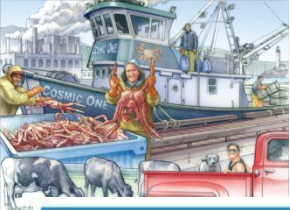
Externalities

- ‘Markets are usually a good way to organize economy activity’
 - In absence of market failures, the competitive market outcome is efficient, maximizes total surplus
- **Externality: one type of market failure**
 - The uncompensated impact of one person’s actions on the well-being of a bystander



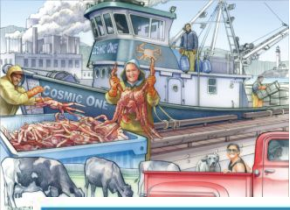
Externalities

- Negative externality
 - Impact on the bystander is adverse
- Positive externality
 - Impact on the bystander is beneficial
- Self-interested buyers and sellers
 - Neglect the external costs or benefits of their actions
 - So the market outcome is not efficient



Externalities

- ‘Government action can sometimes improve upon market outcomes’
 - Why markets sometimes fail to allocate resources efficiently
 - How government policies can potentially improve the market’s allocation
 - What kinds of policies are likely to work best

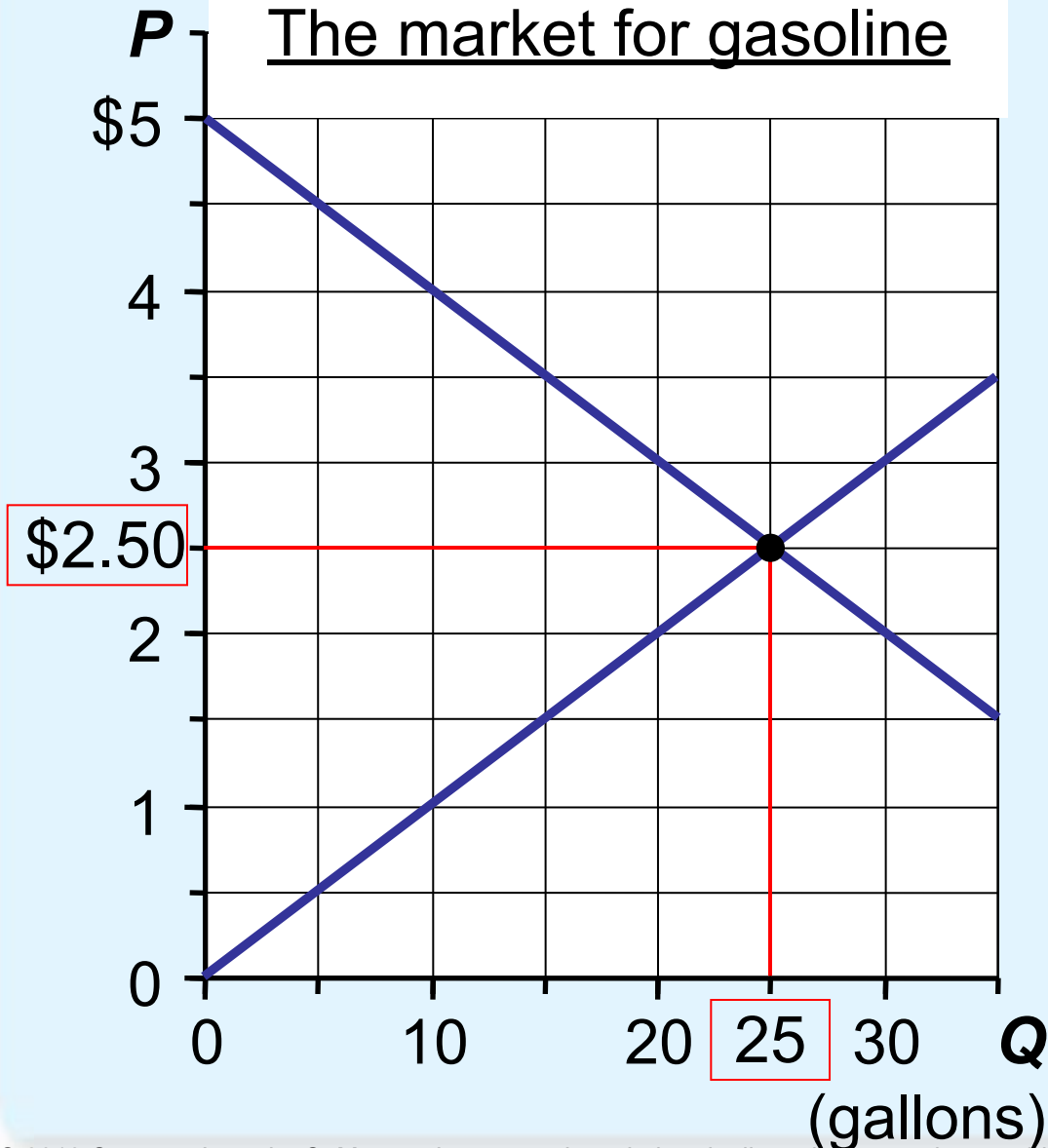


Negative Externalities

- Examples of negative externalities
 - Air pollution from a factory
 - The neighbor's barking dog
 - Late-night stereo blasting from the dorm room next to yours
 - Noise pollution from construction projects
 - Health risk to others from second-hand smoke
 - Talking on cell phone while driving makes the roads less safe for others

Recap of Welfare Economics

The market for gasoline

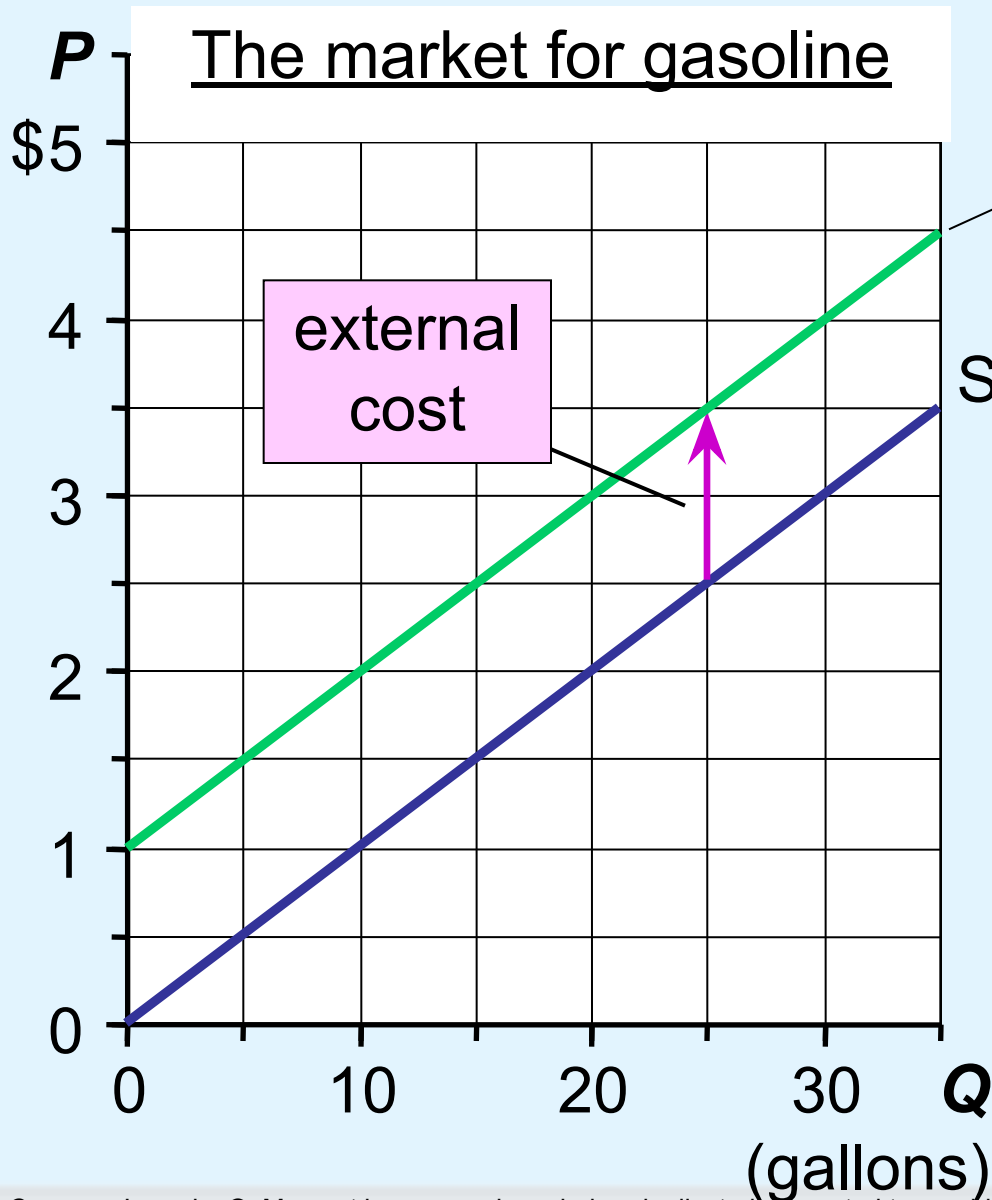


The market equilibrium maximizes consumer + producer surplus.

Supply curve shows **private cost**, the costs directly incurred by sellers.

Demand curve shows **private value**, the value to buyers (the prices they are willing to pay).

Analysis of a Negative Externality



Social cost

= private + external cost

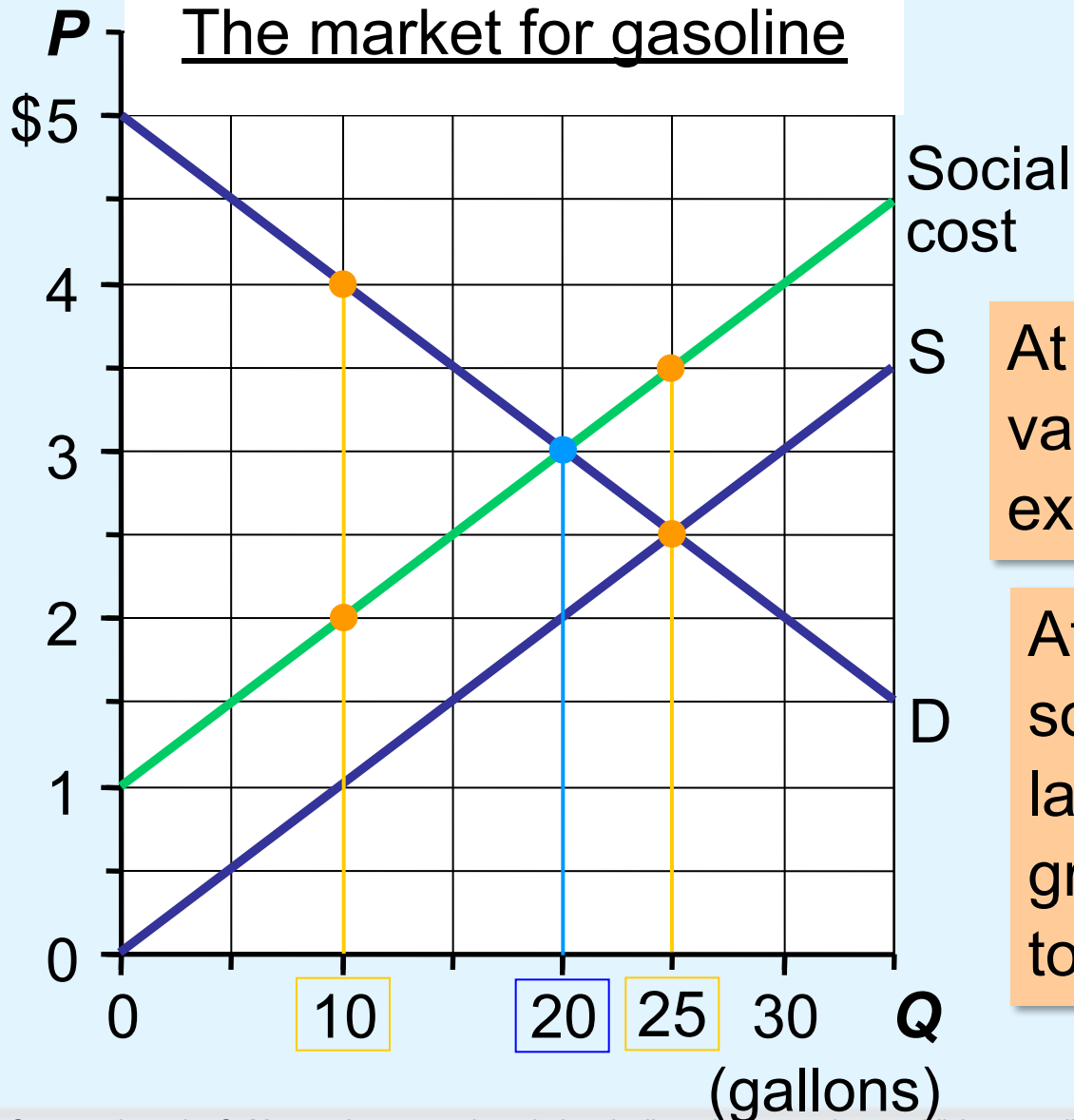
Supply (private cost)

External cost

= value of the
negative impact
on bystanders

= \$1 per gallon
(value of harm
from smog,
greenhouse gases)

Analysis of a Negative Externality

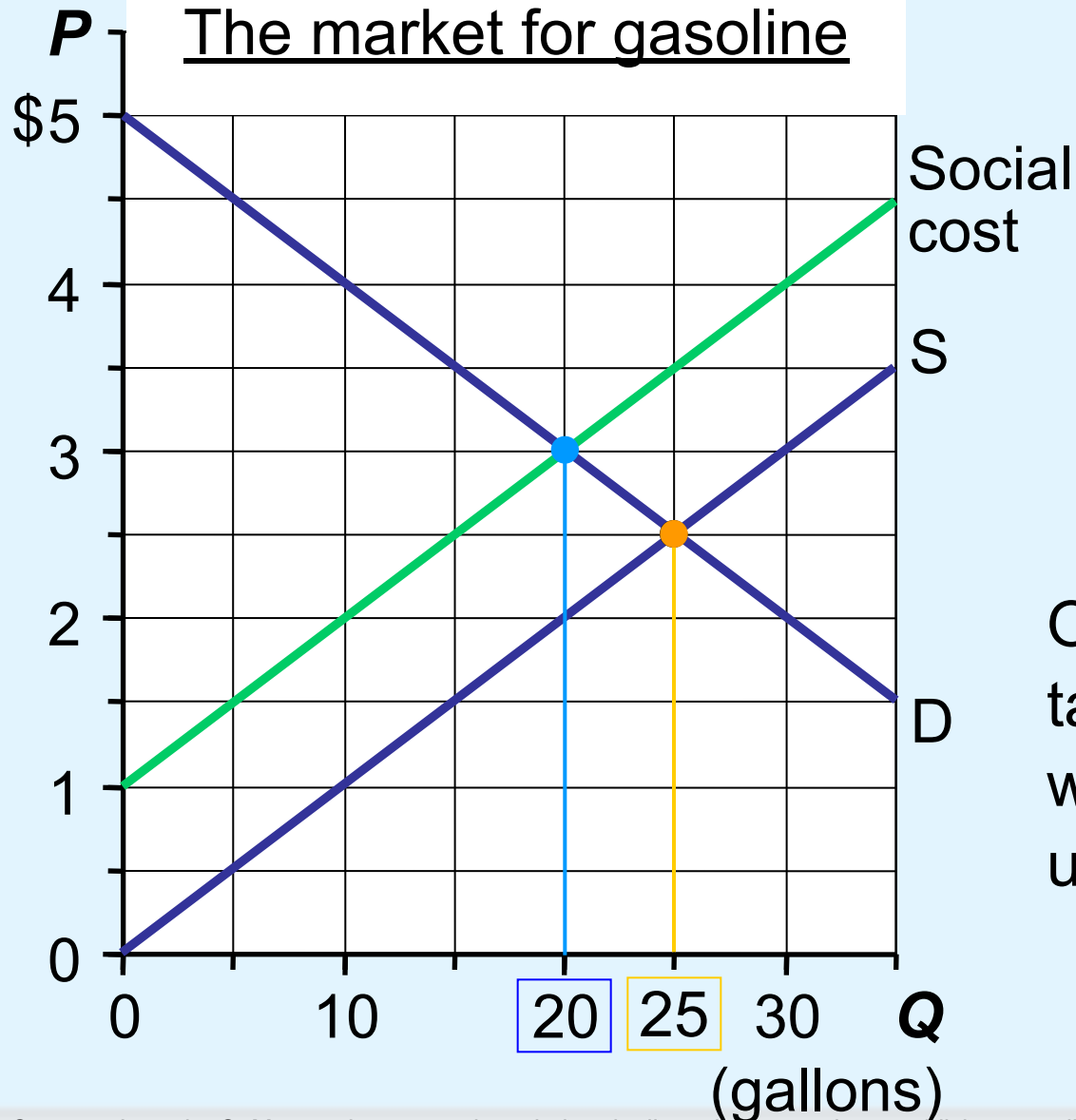


The socially optimal quantity is 20 gallons.

At any $Q < 20$, value of additional gas exceeds social cost.

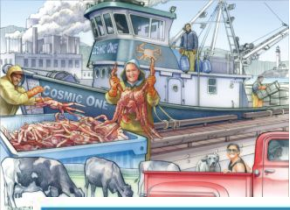
At any $Q > 20$, social cost of the last gallon is greater than its value to society.

Analysis of a Negative Externality



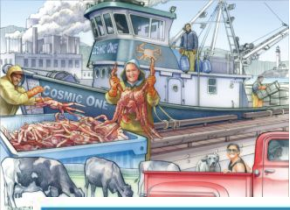
Market equilibrium
($Q = 25$)
is greater than
social optimum
($Q = 20$).

One solution:
tax sellers \$1/gallon,
would shift **S** curve
up \$1.



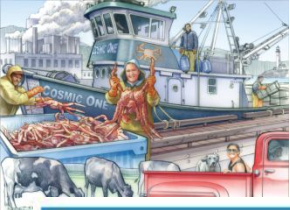
Externalities

- Internalizing the externality:
 - Altering incentives so that people take into account the external effects of their actions
 - In our example, the \$1/gallon tax on sellers makes sellers' costs = social costs.
- If market participants pay social costs
 - Market equilibrium = social optimum



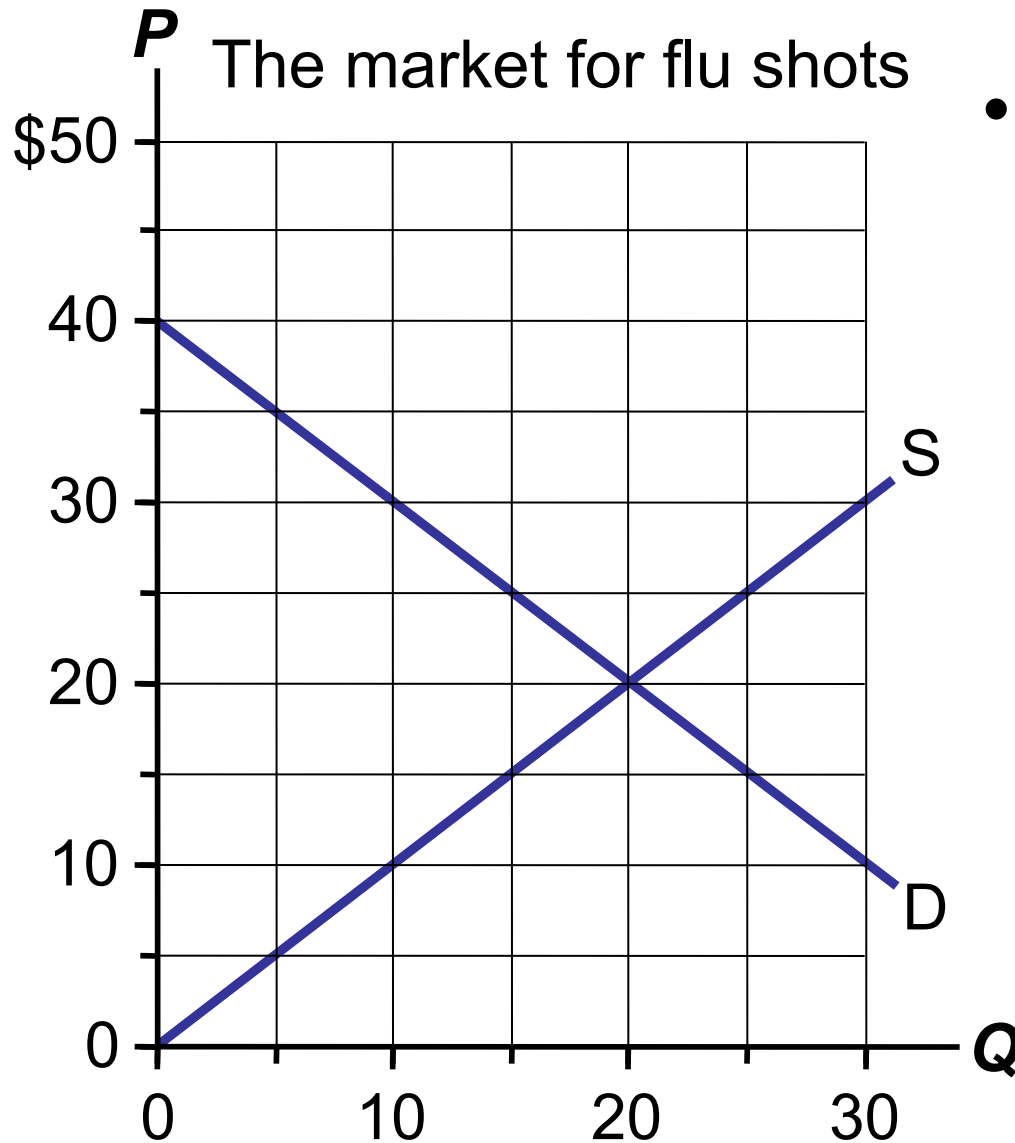
Positive Externalities

- Examples of positive externalities
 - Being vaccinated against contagious diseases protects not only you, but people who visit the salad bar or produce section after you
 - Research and development creates knowledge others can use
 - People going to college raise the population's education level, which reduces crime and improves government

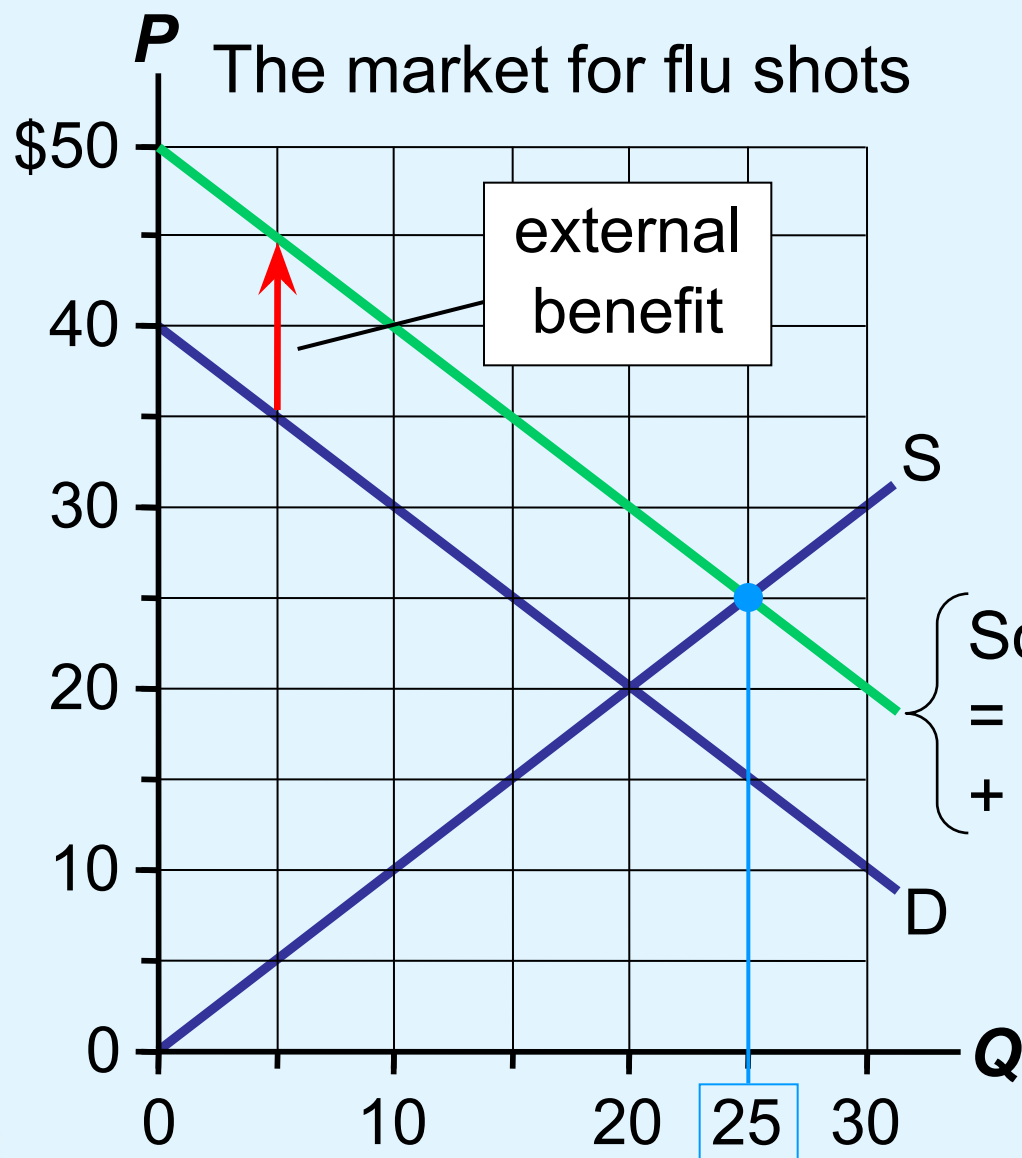


Positive Externalities

- With a positive externality
 - The social value of a good includes
 - Private value – the direct value to buyers
 - External benefit – the value of the positive impact on bystanders
- The socially optimal Q maximizes welfare:
 - At any lower Q , the social value of additional units exceeds their cost.
 - At any higher Q , the cost of the last unit exceeds its social value.



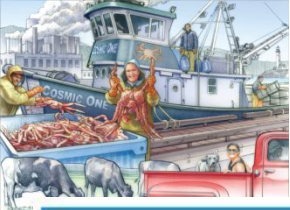
- External benefit = \$10/shot
 - Draw the social value curve.
 - Find the socially optimal Q.
 - What policy would internalize this externality?



Socially optimal Q
= 25 shots.

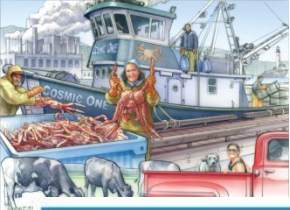
To internalize the
externality, use
subsidy = \$10/shot.

Social value
= private value
+ \$10 external benefit



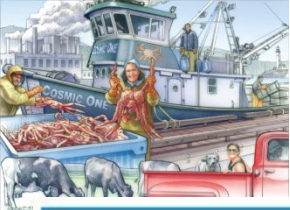
Effects of Externalities: Summary

- If negative externality
 - Market quantity larger than socially desirable
- If positive externality
 - Market quantity smaller than socially desirable
- To remedy the problem, “internalize the externality”
 - Tax goods with negative externalities
 - Subsidize goods with positive externalities



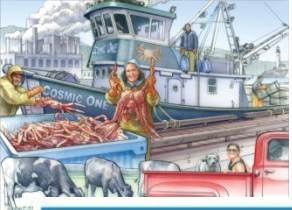
Public Policies Toward Externalities

- **Command-and-control policies**
 - Regulate behavior directly
 - Limits on quantity of pollution emitted
 - Requirements that firms adopt a particular technology to reduce emissions
- **Market-based policies**
 - Incentives so that private decision makers will choose to solve the problem on their own
 - Corrective taxes and subsidies
 - Tradable pollution permits



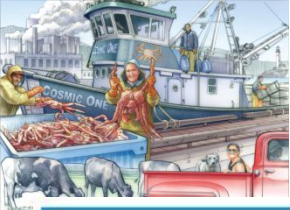
Public Policies Toward Externalities

- Corrective taxes and subsidies
 - Corrective taxes (*Pigovian taxes*)
 - Induce private decision makers to take account of the social costs that arise from a negative externality
 - Places a price on the right to pollute
 - Reduce pollution at a lower cost to society
 - Raise revenue for the government
 - Enhance economic efficiency



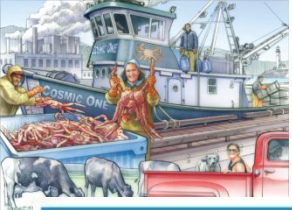
Corrective Taxes vs. Regulations

- Different firms
 - Have different costs of pollution abatement
- Efficient outcome
 - Firms with the lowest abatement costs reduce pollution the most



Corrective Taxes vs. Regulations

- A pollution tax is efficient:
 - Firms with low abatement costs will reduce pollution to reduce their tax burden.
 - Firms with high abatement costs have greater willingness to pay tax.
- Regulation requiring all firms to reduce pollution by a specific amount
 - Is not efficient



Corrective Taxes vs. Regulations

- Corrective taxes are better for the environment:
 - The corrective tax gives firms incentive to continue reducing pollution as long as the cost of doing so is less than the tax
 - If a cleaner technology becomes available, the tax gives firms an incentive to adopt it
 - In contrast, firms have no incentive for further reduction beyond the level specified in a regulation

Example of a Corrective Tax: The Gas Tax

The gas tax targets three negative externalities:

- Congestion

The more you drive, the more you contribute to congestion.

- Accidents

Larger vehicles cause more damage in an accident.

- Pollution

Burning fossil fuels produces greenhouse gases.

Acme and US Electric run coal-burning power plants. Each emits 40 tons of sulfur dioxide per month, total emissions = 80 tons/month.

- Goal: Reduce SO₂ emissions 25%, to 60 tons/month
- Cost of reducing emissions:
 - \$100/ton for Acme
 - \$200/ton for US Electric

- Policy option 1: Regulation
Every firm must cut its emissions 25% (10 tons).
- Your task: Compute the cost to each firm and total cost of achieving goal using this policy.

Each firm must reduce emissions by 10 tons.

Cost of reducing emissions: \$100/ton for Acme, \$200/ton for US Electric.

- Compute cost of achieving goal with this policy:
 - Cost to Acme: $(10 \text{ tons}) \times (\$100/\text{ton}) = \$1000$
 - Cost to USE: $(10 \text{ tons}) \times (\$200/\text{ton}) = \$2000$
 - Total cost of achieving goal = **\$3000**

- Policy option 2: Tradable pollution permits
 - Issue 60 permits, each allows one ton SO₂ emissions.
 - Give 30 permits to each firm.
 - Establish market for trading permits.
 - Each firm may use all its permits to emit 30 tons, may emit < 30 tons and sell leftover permits, or may purchase extra permits to emit > 30 tons.
- Your task: Compute cost of achieving goal if Acme uses 20 permits and sells 10 to USE for \$150 each.

Goal: reduce emissions from 80 to 60 tons

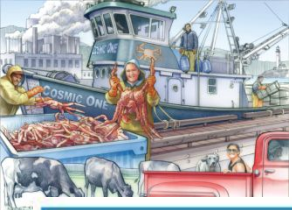
Cost of reducing emissions: \$100/ton for Acme, \$200/ton for USE.

- Compute cost of achieving goal for Acme:
 - Sells 10 permits to USE for \$150 each, gets \$1500
 - Uses 20 permits, emits 20 tons SO₂
 - Spends \$2000 to reduce emissions by 20 tons
 - Net cost to Acme: $\$2000 - \$1500 = \$500$

- Compute cost of achieving goal for USE:
 - Buys 10 permits from acme, spends \$1500
 - Uses these 10 plus original 30 permits, emits 40 tons
 - Spends nothing on abatement
 - Net cost to USE = **\$1500**

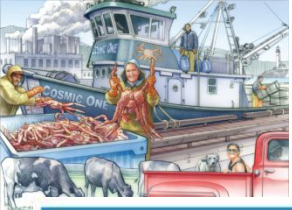
Total cost of achieving goal = \$500 + \$1500 = \$2000

- Using tradable permits, goal is achieved at lower total cost and lower cost to each firm than using regulation.



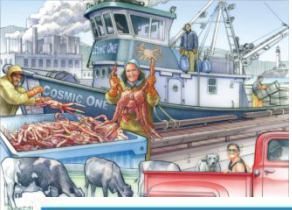
Public Policies Toward Externalities

- Tradable pollution permits system
 - Reduces pollution at lower cost than regulation
 - Firms with low cost of reducing pollution do so and sell their unused permits
 - Firms with high cost of reducing pollution buy permits
 - Result: Pollution reduction is concentrated among those firms with lowest costs



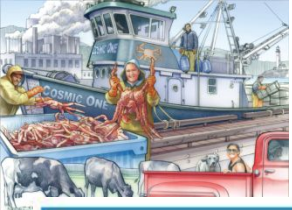
Public Policies Toward Externalities

- Reducing pollution using pollution permits or corrective taxes
 - Firms pay for their pollution
 - Corrective taxes: pay to the government
 - Pollution permits: pay to buy permits
 - Internalize the externality of pollution



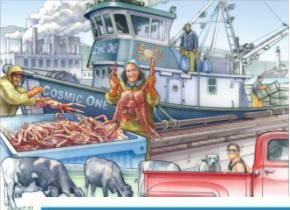
Public Policies Toward Externalities

- Objections to the economic analysis of pollution
 - “We cannot give anyone the option of polluting for a fee.” - by late Senator Edmund Muskie
- People face trade-offs
 - Eliminating all pollution is impossible
 - Clean water and clean air – opportunity cost: lower standard of living



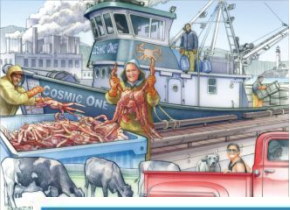
Public Policies Toward Externalities

- Clean environment is a normal good
 - Positive income elasticity
 - Rich countries can afford a cleaner environment
 - More rigorous environmental protection
 - Clean air and clean water - law of demand
 - The lower the price of environmental protection
 - The more the public will want it



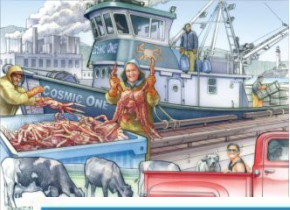
Private Solutions to Externalities

- The types of private solutions
 - Moral codes and social sanctions
 - Charities
 - Self-interest of the relevant parties
 - Integrating different types of businesses
 - Interested parties can enter into a contract



Private Solutions to Externalities

- The Coase theorem
 - If private parties can bargain without cost over the allocation of resources
 - They can solve the problem of externalities on their own
- Whatever the initial distribution of rights
 - Interested parties can reach a bargain:
 - Everyone is better off
 - Outcome is efficient



Private Solutions to Externalities

Why private solutions do not always work

- High transaction costs
 - Costs that parties incur in the process of agreeing to and following through on a bargain
- Stubbornness: bargaining simply breaks down
- Coordination problems
 - Large number of interested parties
 - Coordinating everyone is costly

Summary

- When a transaction between a buyer and seller directly affects a third party, the effect is called an externality.
 - If an activity yields negative externalities, such as pollution, the socially optimal quantity in a market is less than the equilibrium quantity.
 - If an activity yields positive externalities, such as technology spillovers, the socially optimal quantity is greater than the equilibrium quantity.

Summary

- Governments pursue various policies to remedy the inefficiencies caused by externalities.
 - Regulating behavior
 - Internalizes an externality using corrective taxes
 - Issue permits (similar results to imposing corrective taxes on polluters)

Summary

- According to the Coase theorem, if people can bargain without cost, then they can always reach an agreement in which resources are allocated efficiently.
 - In many cases, however, reaching a bargain among the many interested parties is difficult, so the Coase theorem does not apply.